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Context Update with Cantonese Sentence-Final Particle *me1* and the Role of the Final Low Tone

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1. Introduction

Cantonese has a large inventory of sentence-final particles (SFPs), at least 40 distinct forms by some estimates (Sybesma & Li 2007), which serve a range of pragmatic and discourse functions, including marking speech-act types, managing common ground, and signaling the speaker's epistemic stance (Chor 2018, Matthews & Yip 2010). Among these, the biased-question particle *me1* has drawn particular attention for its link to the speaker's negative stance (Matthews & Yip 2010, Chor & Lam 2023, Hara 2023, Law et al. 2024). For example, in (1), *me1* conveys that the speaker has a negative bias towards its prejacent proposition while seeking confirmation in response to some unexpected evidence.

(1) [*Context: Ming has been working hard on his paper that is due very soon. His roommate Gyun sees that he is lying on the couch doing nothing. Gyun asks:*]

Nei5 leon6man2 se2-jyun4 laa3 **me1**?
your paper write-finish LAA3 *me1* 'You finished writing your paper?'

Additionally, there is a high-falling tonal variant (henceforth *me*↓)[1] that has been described as the rhetorical use of *me1* (Matthews & Yip 2010), comparable to English matrix *as if* (Rett & Starr 2023). As shown in (2), *me*↓ is not used to request information or seek confirmation, but to express disbelief or rejection towards its prejacent.

(2) [*Context: Ming has been working on his paper. His friend said he should take a rest during spring break. Ming says:*]

Leon6man2 sik1 zi6gei2 se2-jyun4 **me**↓
paper know self write-finish ME↓ 'As if the paper could finish writing itself.'

The paper offers a formal analysis of the SFP *me1* and its tonal variant *me*↓ within the Table Model (Farkas & Bruce 2010, Farkas & Roelofsen 2017). §2 presents the core empirical generalizations about *me1*. §3 develops the analysis. §4 sketches a preliminary account of *me*↓, which I propose to be compositionally derived from the base SFP *me1* in combination with a final low-tone particle. §5 concludes the paper.

2. The Bias Profile of *me1*

Previous work described the Cantonese SFP *me1* as marking a negative epistemic bias toward its prejacent proposition, often formalized as a presupposition of speaker disbelief or skepticism (Sybesma &

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[1] Standard Cantonese has six lexical tones (Bauer & Benedict 1997): High Level (55, 53; T1), Mid Rising (25; T2), Mid Level (33; T3), Low Falling (21; T4), Low Rising (23; T5), and Low Level (22; T6). The high-falling contour on *me*↓ (51) is not among them. Examples in this paper are romanized in Jyutping using these tone labels.

Li 2007, Matthews & Yip 2010, Chor & Lam 2023, Law et al. 2024). For example, Chor & Lam (2023) analyze *me1* as a negative stance marker for which "a 'no' answer is preferred by the speaker", where they characterize (3) as the speaker's skepticism that the addressee would go hiking.

(3) [*Context: A person challenges his friend who is very fond of hiking, questioning him whether the friend would go hiking in such hot weather*] (Chor & Lam 2023: 76)

Gam3 jit6, nei5 wui3 heoi3 me1?
so hot you will go ME1 'It is so hot – you going?'

I show that treating *me1* solely as encoding a negative bias toward p cannot capture the full range of data that *me1* is compatible with. In §2.1, I argue that the "negative presupposition" is better analyzed as a prior belief of the speaker; crucially, at the time of utterance, the speaker's bias towards p can vary. In §2.2, I show that *p-me1* questions require that the speaker has reason to assume that the addressee may believe p to be true, a characteristic not predicted by a purely speaker-oriented bias account.

2.1. Variable current bias and invariable prior negative belief

As noted above, traditional analyses of *me1* characterize it as encoding a negative presupposition, such that a *p-me1* question communicates the speaker's epistemic bias towards $\neg p$ (Chor & Lam 2023, Chor 2018, Sybesma & Li 2007, Matthews & Yip 2010, Law et al. 2024, Lam 2014). This assumption captures cases like (3), where the speaker communicates skepticism toward the prejacent proposition, and extends to stronger cases where the speaker holds a clear belief in $\neg p$. In (4), for instance, the speaker clearly believes that *two plus three does NOT equal six*.

(4) [*Context: Gyun is tutoring Manzai, a first grader. Manzai wrote down 2+3=6. Gyun asks:*]

Ji6 gaa1 saam1 dang2jyu1 luk6 me1?
two plus three equal six ME1 'Two plus three equals six?

However, closer examination shows that the interpretation of *me1* accommodates a much wider range of current speaker bias than traditionally assumed. Specifically, while the speaker signals that the prejacent proposition is surprising or unexpected, their current epistemic stance toward the prejacent at the time of utterance can vary depending on the context. For instance, in (5), the speaker conveys a suspicion that the prejacent (i.e., *Gyun drank milk*) may be true despite it being counter-expectational, rather than communicating a negative bias toward it. In (6), the speaker appears to hold a strong positive bias toward the prejacent (i.e., *Ming eats meat*), despite the prior assumption that the addressee was vegan.

(5) [*Context: Gyun is lactose intolerant, Ming hears that her stomach is growling. He asks:*]

Nei5 jam2-zo2 naai5 **me1**?
you drink-PERF milk ME1 'You drank milk?'

(6) [*Context: Gyun thought Ming was vegan. Ming just ordered a chicken taco. Gyun asks:*]

Haa2? Nei5 sik6 juk6 ge3 me1?
huh you eat meat GE3 ME1 'Huh? You eat meat?'

Examples like (5) and (6) show that while *me1* marks a departure from prior expectations, it does not rigidly encode a current negative bias. It is also compatible with a positive current bias, even as strong as a belief in p. Nonetheless, native speaker intuitions consistently associate *me1* with some form of negativity. For instance, in Hara's (2023) rating study, *me1*-questions were judged most natural in negative contexts.[2]

I take these intuitions to be correct: *me1* carries a negative component, but not necessarily as the speaker's current stance. Rather, following Wakefield (2014), I argue that *me1* reflects a prior belief in $\neg p$ that is now being reconsidered in light of contextual evidence.[3] On this view, *me1* signals that p is counter to prior expectations, even if the speaker is currently open to or convinced of p.

[2] Note that in Hara (2023), the stimuli did not differentiate between prior and current bias.

[3] I follow Goodhue's (2023) characterization of contextual evidence as mutually available evidence in the discourse

Evidence for the prior belief requirement comes from cases like (7), which are infelicitous when the speaker lacks any relevant prior expectation about p.[4]

(7) [*Context: Ming went on a date with Gyun for the first time. Ming had no idea about Gyun's preferences, so he got her two bundles of flowers: a bundle of roses and a bundle of tulips. Gyun's first reaction was to take the roses.*]

??Nei5	zung1ji3	mui4gwai3faa1	do1	gwo3	wat1gam1hoeng1	*me1*?
you	like	rose	more than		tulip	ME1

Intended: 'You like roses more than tulips?'

In sum, I follow prior work in treating *me1* as encoding a negative component, but I argue that this does not amount to an invariant bias against p at the time of utterance. Rather, *me1* reflects a discourse-sensitive revision of a prior belief in $\neg p$, prompted by contextual evidence that makes p a relevant possibility, with the speaker's current bias toward p varying depending on the context.

2.2. *Invariable assumption of addressee belief*

Previous analyses of *me1* have focused primarily on speaker-oriented epistemic bias, typically characterizing it as encoding the speaker's skepticism or disbelief toward the prejacent proposition. To my knowledge, no prior work has examined the inferences *me1* licenses about the addressee's epistemic state. I argue that a felicitous *p-me1* question requires that the speaker has reasons to assume that the addressee may believe p to be true, a condition that cannot be reduced to speaker bias alone.

In (8), the *p-me1* question is felicitous in *Context 1*, where the speaker can assume that the addressee may believe *durian is delicious*. In *Context 2*, however, the speaker cannot assume so, rendering the utterance infelicitous.

(8) [*Context 1: Abe and Ben had never had durian before. They were at a durian-tasting party. Ben had his 10^th^ piece while Abe was still too afraid to try. Abe asks Ben:*]
[*#Context 2: ...Ben had a try and stopped while all the others kept devouring theirs. Abe asks Ben:*]

Hou2sik6 **me1**?
delicious ME1 'It is delicious?'

The requirement that the speaker invariably assumes the addressee's belief in p is further illustrated by (9), in which the addressee holds epistemic authority, making it inappropriate for the speaker to openly presuppose their beliefs. In this context, uttering the *me1*-question in (9) would be perceived as impolite, as it conveys the assumption that the professor may genuinely believe the incorrect equation.

(9) [*Context: Gyun's professor wrote 2+3=6 on the board. She points at the equation and asks:*]

??Hai6 luk6 me1?
COP six ME1 Intended: 'It is six?'

3. Analysis

This section develops a formal analysis of *me1* within the Table Model (Farkas & Bruce 2010, Farkas & Roelofsen 2017), accounting for its bias profile. §3.1 introduces the Table Model of discourse

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situation, which, disregarding conflicting beliefs, would allow participants to infer p or at least increase the likelihood of inferring p. For instance, in the $2 + 3 = 6$ example in (4), the fact that the student wrote down the wrong equation increases the likelihood of inferring p (i.e., *two plus three equals six*) in the discourse context, despite it conflicting with the speaker's belief.

[4] Note that for (7), the consultants did not judge it to be fully infelicitous. This is due to the fact that the participants could still force a presupposition of negative belief. Four native speakers indicated that they obligatorily get an interpretation that Ming had an assumption that people would prefer tulips in general, i.e., a prior negative belief, which is in line with the current analysis.

and its application. §3.2 analyzes *me1* as a lexically specified utterance function within this framework.

3.1. The Table Model

I adopt the Table Model advanced by Farkas & Bruce (2010) and elaborated by Farkas & Roelofsen (2017), which analyzes the meaning of utterances in terms of their effects on an evolving conversational context. It analyzes utterances in a conversation between a set of interlocutors A as functions from an input context to an output context. The model assumes the components in (10).

(10) a. Discourse Commitments (DC): for each discourse participant $a \in A$, DC_a contains propositions that a has publicly committed to
b. Common Ground (CG): the set of all propositions that all discourse participants are publicly committed to
c. The Table: a stack of Issues I (sets of propositions) to be resolved by common-grounding one of their members
d. Projected Set (PS): the set of all potential future CGs, containing $CG_i + p$ for each $p \in I$ that is compatible with the beliefs of all interlocutors.

The architecture of the model conceptualizes conversation as a cooperative process oriented toward Common Ground growth, reducing uncertainty about the shared state of the world by negotiating and resolving Issues. Each discourse move proposes potential Common Ground updates. The addressee may then respond by committing to a $p \in I$, challenging the Issue (e.g., by introducing incompatible content), or withholding resolution when there is no $p \in I$ that is currently mutually acceptable. Each type of response promotes Common Ground growth either by adding a proposition to it or by constraining the space of its viable future states.

Farkas & Roelofsen (2017) formalize the conventional utterance function from an input context to an output context as in (11). The utterance function utt takes as arguments a set of propositions P of type $< st, t >$, a speaker Sp of type e, and an input context c of type k, and returns an output context. For an utterance denoting a set of propositions P by a speaker Sp, the speaker puts P on the table, and publicly commits to $\bigcup P$, and projects $CG_i + p$ for each $p \in P$ as potential future common grounds.

(11) $$[\![\textsc{utt}]\!] = \lambda P_{(st)t}.\lambda Sp_e.\lambda c_k. \begin{bmatrix} T & = & T_c + P \\ DC_{Sp} & = & DC_{Sp,c} + \bigcup P \\ PS & = & CG_c + p : p \in P \\ c' & = & c_c \text{ in all other respects} \end{bmatrix}^{c'}$$

To illustrate, consider how the model derives the context update effects of two basic sentence types. The context update of a canonical declarative sentence that denotes a singleton proposition p is illustrated in (12), and that of a canonical polar question denoting $\{p, \neg p\}$ in (13).

(12) Sp : *Linguistics is fun.*
a. The set $P = \{p\}$ (where p = 'linguistics is fun') is placed on the Table
b. $\bigcup P = p$ is added to DC_{Sp}
c. Sp presents p as a candidate addition to the CG, the addressee Ad may accept (by adding p to DC_{Ad}), reject (e.g., by introducing $\neg p$), or withhold.

(13) Sp : *Is linguistics fun?*
a. The set $P = \{p, \neg p\}$ (where p = 'linguistics is fun',$\neg p$ = 'linguistics is not fun') is placed on the Table
b. $\bigcup P = \bigcup\{p, \neg p\} = U$ is added to DC_{Sp}. Since $DC_{Sp} + U = DC_{Sp}$, Sp is essentially making no new commitments.
c. Sp presents p and $\neg p$ as candidate additions to the CG, the addressee Ad may accept (by adding p or $\neg p$ to DC_{Ad}) or withhold from resolution.

The canonical cases above provide the baseline against which bias-marked utterances can be compared. Non-canonical types depart from these baseline updates by combining the conventional structure with features that alter the speaker's commitment profile. English rising declaratives (RDs) such as '*Linguistics is fun?*' are a well-discussed example, commonly analyzed as the rising tune (L* H-H%) modifying the context-update function of declarative sentences (Gunlogson, 2001; Westera, 2017; Truckenbrodt, 2012; Rudin, 2022, 2018; Goodhue, to appear). Their bias profile is characterized by variable speaker bias toward p, ranging from negative to positive, together with an invariable assumption that the addressee may believe p (see detailed discussion in Rudin, 2022). Following Rudin (2022), I assume that the rising tune modifies the utterance function by removing the speaker's discourse commitment to $\bigcup P$ (see also Truckenbrodt, 2012), as in (14). The context update of an English RD is illustrated in (15).

(14) $$[\![\text{L* H-H\%}]\!] = \lambda K_{kk}.\lambda Sp_e.\lambda c_k.\begin{bmatrix} DC_{Sp} & = & DC_{Sp,c} \\ c' & = & K(sp,c) \text{ in all other respects} \end{bmatrix}^{c'}$$

(Adapted from Rudin, 2022:359)

(15) Sp : *Linguistics is fun?* (*Linguistics is fun* + L* H-H%)

a. The set $P = \{p\}$ (where p = 'linguistics is fun') is placed on the Table
b. Sp makes no commitment regarding P
c. Sp presents p as a candidate addition to the CG, the addressee Ad may accept (by adding p to DC_{Ad}), reject (e.g., by introducing $\neg p$), or withhold.

What does it mean for a speaker to raise p, signal that it may be added to the future CG, yet refrain from committing to its truth? In RDs, this configuration arises because the rising tune refrains the speaker from committing to p while still projecting p. Rudin (2022) argues that for this move to make sense, the speaker must treat p as a live option in the conversation. The only way to make this move cooperative is that the speaker is uncertain about p but thinks it is possible that the addressee believes p. If the speaker believed p to be true, they would commit to it; if the speaker believed p to be false, projecting p would be misleading. Crucially, since the speaker projects only p but not $\neg p$, this only makes sense if the speaker assumes that only p is compatible with the addressee's belief (and not $\neg p$). This explains the stable addressee-bias inference and the variable speaker-bias profile: RDs can occur when the speaker leans towards p, against p, or is agnostic, as long as they find it plausible that the addressee believes p (see detailed pragmatic reasoning in Rudin, 2022).

3.2. *Me1 as a lexically specified utterance function*

Taking inspiration from English RDs, I propose a formal compositional analysis of *me1* as an utterance function operator[5] that patterns with English RDs in its discourse effects: a *p-me1* utterance places the prejacent proposition p on the Table, signals that $CG + p$ is a viable future update, and crucially, withholds any public speaker commitment to p. Unlike English RDs, however, *me1* carries an additional presupposition of the speaker's prior negative belief. This contrast is illustrated in (16), where there is contextual evidence for p but the speaker has no prior negative belief. The English RD in (16-a) is felicitous in this context, whereas the Cantonese *me1*-counterpart does not survive. I treat this presupposition as part of *me1*'s lexical meaning.

(16) [*Context: Rose walks into the kitchen and sees that the oven is on and the smell of freshly baked cookies is in the air. She asks her roommate:*]

[5] Note that for Rudin (2022), the meaning of an English RDs arises from the combination of a general utterance function (in the sense of Farkas & Roelofsen (2017) and the rising tune as a modifier of that function. In contrast, I do not treat *me1* as an utterance function modifier on par with the English rising tune, but rather as an utterance function in and of itself. That is, *me1*-marked utterances directly instantiate dynamic context update functions, without requiring an additional layer of a general-purpose utterance operator. While positing such a general utterance function might be independently motivated in other frameworks (Farkas & Roelofsen 2017, Gunlogson 2001, Roberts & Rudin 2024), it does not make a crucial difference for the present analysis. In fact, this simplification is empirically grounded: in Cantonese, utterances without any SFP are generally dispreferred in natural conversations, suggesting that these particles are responsible for introducing the update potential of the sentence.

a. (English RD:) You baked cookies?
b. (Cantonese *me1*:)

??Nei5 guk6-zo2 kuk1kei4 me1?
you bake-PERF cookie ME1 Intended: 'You baked cookies?'

The proposed denotation of *me1* is given in (17).

(17) $$[\![\text{ME}1]\!] = \lambda p_{st}.\lambda Sp_e.\lambda c_k. \begin{bmatrix} T & = & T_c + \{p\} \\ PS & = & \{CG_c + p\} \\ c' & = & c_c \text{ in all other respects} \end{bmatrix}^{c'}$$
defined iff $\exists t'.\exists t.\ \texttt{believe}(Sp, \neg p, t') \wedge \texttt{ContEvi}(p, t) : t' < t$
where $\texttt{ContEvi}(p, t)$ is true iff contextual evidence for p is available at t

A *p-me1* question like (1) achieves the discourse effect as in (18).

(18) Sp : *You finished writing your paper + me1*? (=(1))
a. The set $P = \{p\}$ (where p = 'you finished writing your paper') is placed on the Table
b. Sp makes no commitment regarding P
c. Sp signals that they held a prior belief in $\neg p$ before encountering some discourse evidence
d. Sp presents p as a candidate addition to the CG, the addressee Ad may accept (by adding p to DC_{Ad}), reject (e.g., by introducing $\neg p$), or withhold.

Cantonese *me1* gives rise to the same core pragmatic inference as English RDs: the speaker's stance toward p can vary at the time of utterance, but it consistently conveys the speaker's assumption that the addressee may believe p. The additional presupposition of prior negative belief further constrains its use, distinguishing *me1* from RDs while preserving the shared bias profile.

4. A Compositional Analysis of *me*↓

The *SFP me1* may appear with a high-falling tone, giving rise to a rhetorical interpretation similar to the English matrix "as if" (Rett & Starr 2023). In a *p-me*↓ utterance, the speaker expresses disbelief toward the prejacent proposition p, while acknowledging that contextual evidence makes p superficially plausible. For example, in (19), the speaker does not believe that *the paper could finish writing itself*, but the advice made by the addressee seemingly suggests its superficial possibility.

(19) [*Context: Ming has been working on his paper. His friend said he should take a rest during spring break. Ming says:*] (=(2))

Leon6man2 sik1 zi6gei2 se2-jyun4 **me**↓
paper know self write-finish ME↓ 'As if the paper could finish writing itself.'

Previous studies have documented systematic semantic contrasts across tonal variants of Cantonese SFPs. For instance, Matthews & Yip (2010) observe that high-tone variants typically convey tentativeness, while low-tone variants are often associated with assertiveness. Efforts have also been made to formalize the semantic contributions of tonal morphemes on SFPs (Law 1990, Cheung 1986, Sybesma & Li 2007). For instance, Sybesma & Li (2007) argue that tone itself can serve as a minimal semantic unit on SFPs. Additionally, recent phonetic studies have revealed that the Cantonese low-tone SFPs arise from the combination of an inherent lexical tone and an additional low boundary tone (Lee 2021, Lee et al. 2024), providing independent evidence in support of a semantically compositional analysis. Building on the previous works, I propose that *me*↓ is compositionally derived from the combination of the SFP *me1* and a final low-tone particle,[6] as illustrated in (20). The combination of the high-level tone on *me1* and the final low tone is phonetically realized as the high-falling tone on *me*↓.

[6] In this analysis, the final low tone shares a similar status as SFPs. E.g., Chao (1968) suggests that final rising and falling intonations (in Mandarin Chinese) should be treated as distinct SFPs; see also Wakefield (2016), who has argued that sentence-final intonation and SFPs are "the same thing in different form" in Cantonese.

(20) [[[p] *me1*] %L]

I propose that the final low-tone particle in *me*↓ contributes an independent semantic component. Specifically, it encodes that the prejacent proposition p is incompatible with some interlocutor's discourse commitments in the input context, formalized in (21).

(21) $[\![\text{L\%}]\!] = \lambda \textsc{utt}.\textsc{utt}$ defined iff $\exists a \in A$ such that $DC_{a,i} \cup p \vdash \bot$

Following Farkas (to appear), I assume that this inconsistency blocks the projection of p, as also defined as a requirement for *PS* in (10). In other words, a *p-me*↓ utterance is identical to a *p-me1* utterance except that *p-me*↓ does not project $CG + p$ as a viable path forward. A *p-me*↓ utterance like (2) achieves the discourse effect as in (22), where it does not lead toward any modification to the current CG.

(22) Sp : *The paper could write itself* + *me*↓? (=(2))

a. The set $P = \{p\}$ (where p = 'the paper could write itself') is placed on the Table
b. Sp makes no commitment regarding P, but signals that it is not compatible with some interlocutor's belief
c. Sp signals that they held a $\neg p$ believe prior to encountering some discourse evidence
d. Sp does not project p as a candidate addition to the CG

Building on Farkas (to appear), I propose that the rhetoricality of *me*↓ arises precisely from the non-projection of p. (cf. Sadock 1971, Han 2002, Rohde 2006, Caponigro & Sprouse 2007, Biezma & Rawlins 2017, Rohde to appear). Farkas (to appear) argues that an utterance is understood to be rhetorical if the Issue it raises is either already resolved or is unresolvable, i.e., if it fails to project a *PS* distinct from the current CG. This precisely characterizes *p-me*↓: Sp's non-commitment to p in combination with the inconsistency of p with some interlocutor's commitments prohibits the projection of p.

In this compositional account, the meaning of *me*↓ is derived from the interaction between *me1* and the final low tone. This approach makes predictions about a broader range of SFPs with final low tones. For example, Matthews (1998) argues that the low-falling tone SFP *wo4* is derived from the mid-tone SFP *wo3*, with the low tone linked to surprisal, as illustrated in (23). I propose that *wo4*'s surprisal flavor stems from the final low tone signaling the inconsistency in addition to *wo3*'s base meaning of noteworthiness, where the prejacent proposition, once inconsistent with DC_{Sp}, is now committed to by Sp, hence generating a sense of surprise.

(23) Keoi5 gam1jat6 mou5 faan1hok6 **wo3/wo4**!
he today not.have attend.school wo3/wo4
'(*(Wo3-)*Noteworthily,*(Wo4-)*I didn't expect that) he didn't go to school today!'

Formalizing the compositional interactions between SFPs and final low tones lays the foundation for future research on predictive and explanatory accounts of SFP-tone pairs, such as *aa3* vs. *aa4* and *tim1* vs. *tim*↓, with low-tone variants often tied to surprise or prior negative belief.

5. Summary

This paper has argued that the Cantonese SFP *me1* shares key discourse effects with English RDs: raising p as an Issue in light of some contextual evidence, projecting p as a candidate for inclusion in future Common Ground, and withholding public speaker commitment, while adding a lexical presupposition of the speaker's prior belief in $\neg p$. I showed that *me1* allows variable current speaker bias but invariably conveys that the speaker takes it plausible that the addressee believes p. Its tonal variant *me*↓ was analyzed as compositionally derived from *me1* plus a final low-tone particle encoding the incompatibility of p with some interlocutor's commitments, blocking projection and yielding a rhetorical reading. This account captures the empirical contrasts between *me1* and *me*↓, links them to broader patterns in bias-marked utterances cross-linguistically, and lays the groundwork for extending the compositional tone-based analysis to other Cantonese SFPs.

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